

Compact Groups Give an Arbitrary-Weight Partial Answer to the Dugundji Compact Question

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Source Question

Banach and Kubis ask in arXiv:0610795 whether, for a Dugundji compact space K of weight $> \aleph_1$, the Banach space $C(K)$ must be 1-Plichko. They prove the affirmative result at weight $\leq \aleph_1$ and note that compact groups are Dugundji compacta.

Result

Theorem 1. *Let G be a compact Hausdorff group. Then $C(G)$ has a commutative norm-one projectional skeleton. In particular, $C(G)$ is 1-Plichko.*

Idea of the Proof

Use the group structure to replace the ordinal inverse-sequence argument in the source paper. For every closed normal subgroup N such that G/N is metrizable, Haar averaging over N is a norm-one projection of $C(G)$ onto the functions factoring through G/N . These projections commute because convolution of Haar probability measures on compact normal subgroups N and M is Haar measure on NM . Peter–Weyl supplies density, since finite-dimensional matrix coefficients factor through metrizable quotients.

Proof

Let Γ be the family of all closed normal subgroups $N \triangleleft G$ such that G/N is metrizable. Order Γ by reverse inclusion:

$$N \preceq M \iff N \supseteq M.$$

This is directed because $N \cap M \in \Gamma$, as $G/(N \cap M)$ embeds as a closed subgroup of the metrizable compact group $G/N \times G/M$.

For $N \in \Gamma$, let m_N denote normalized Haar measure on N , and define

$$(P_N f)(g) = \int_N f(gn) dm_N(n) \quad (f \in C(G), g \in G).$$

Then P_N is a contractive projection. It has norm one unless G is empty, and its range is exactly the subspace of functions constant on right cosets of N , i.e. the canonical copy of $C(G/N)$. Thus every range is separable.

If $N \supseteq M$, then every N -coset-constant function is M -coset-constant, and direct averaging gives

$$P_N P_M = P_N = P_M P_N.$$

For arbitrary $N, M \in \Gamma$, the convolution $m_N * m_M$ is the normalized Haar measure on the compact subgroup NM . Indeed it is supported on NM , and because N and M are normal it is left invariant under both N and M , hence under the subgroup generated by them, namely NM . Therefore

$$P_N P_M = P_{NM} = P_M P_N,$$

so the projections commute.

Now let $N_1 \supseteq N_2 \supseteq \dots$ be an increasing sequence in Γ , equivalently $N_1 \supseteq N_2 \supseteq \dots$, and set $N = \bigcap_j N_j$. Then G/N embeds into $\prod_j G/N_j$, hence is metrizable, so $N \in \Gamma$. The quotient maps $G/N \rightarrow G/N_j$ separate points. By the complex Stone–Weierstrass theorem, the union of the pulled back algebras $\bigcup_j C(G/N_j)$ is dense in $C(G/N)$. In terms of the projections, this says

$$P_N[C(G)] = \overline{\bigcup_j P_{N_j}[C(G)]}.$$

This is the required countable-supremum axiom for a projectional skeleton.

It remains to check that the union of all ranges is dense in $C(G)$. By the Peter–Weyl theorem, finite linear combinations of matrix coefficients of finite-dimensional continuous unitary representations of G are uniformly dense in $C(G)$. If $\pi : G \rightarrow U(n)$ is such a representation, then every matrix coefficient factors through $G/\ker \pi$, and $G/\ker \pi$ is metrizable because it is isomorphic to the compact subgroup $\pi(G) \subseteq U(n)$. Hence every matrix coefficient lies in $P_{\ker \pi}[C(G)]$ for some $\ker \pi \in \Gamma$. Thus $\bigcup_{N \in \Gamma} P_N[C(G)]$ is dense.

The family $(P_N)_{N \in \Gamma}$ is therefore a commutative norm-one projectional skeleton on $C(G)$. By the standard characterization of 1-Plichko spaces as Banach spaces with a commutative norm-one projectional skeleton, $C(G)$ is 1-Plichko.

Scope and Novelty Check

The result covers compact groups of arbitrary weight, hence an important Dugundji compact subclass, but it does not settle the source question for all Dugundji compacta.

Local run indexes and parsed sources were searched for compact-group Plichko variants. The local hits record the source paper’s weight $\aleph_1$ compact-group application and later references saying that $C(K)$ is 1-Plichko for Abelian compact groups. I found no existing run packet or local source hit stating the above all-compact-group projectional-skeleton argument.

References

- T. Banach and W. Kubis, *Spaces of continuous functions over Dugundji compacta*, arXiv:0610795.
- W. Kubis, *Banach spaces with projectional skeletons*, J. Math. Anal. Appl. 350 (2009), 758–776. Used for the projectional-skeleton characterization of Plichko spaces as quoted in local later sources.
- The Peter–Weyl theorem for compact groups.

Human Review Recommendation

Check the nonabelian averaging step $P_N P_M = P_{NM}$, specifically the claim that $m_N * m_M$ is Haar measure on NM for compact normal subgroups. This is the only group-theoretic point where commutativity of the group is not available.